Unit – V
Sequencing Problems

Introduction, Notations, Terminology and Assumptions, Processing *n* Jobs Through Two Machines, Processing *n* Jobs Through Three Machines, Processing *n* Jobs Through *m* Machines, Processing Two Jobs Through m Machines

Introduction

The optimal order (sequence) shows the minimum time in which jobs, equipment, people, materials, facilities and all other resources are arranged to support the production schedules to give low costs and high utilizations. Other objectives of calculating optimal production schedule are minimizing customers waiting time for a product or service, meeting promised delivery dates, keeping stock levels low providing preferred working pattern, and so on.

If n jobs are to be performed, one at a time, on each of m machines, where sequence (order) of the machines in which each job should be performed, and the actual (or expected) time required by the jobs on each of the machines are given, then the general sequencing problem is to find a sequence out of $(n!)^m$ possible sequences, which minimize the total elapsed time between the start of the job on first machine and the completion of the last job on the last machine.

In particular, if there are n = 3 jobs to be performed and m = 3 machines are to be used, then the total number of possible sequences will be $(3!)^3 = 216$. Theoretically, it may be possible to find the optimum sequence but this would require a lot of computational time. Thus, one should adopt the sequencing technique.

Notations

t_{ij} = Processing time (time required) for job i on machine j.
T = Total elapsed time for processing all the jobs. This includes idle time, if any.
I_{ij} = Idle time on machine j from the end of job (i – 1) to the start of job i.

Terminology

- *Number of Machines* The number of machines refer to the number of service facilities through which a job must pass before it is assumed to be completed.
- *Processing Time* This is the time required by a job on each machine.
- *Processing Order* This refers to the order (sequence) in which machines are required for completing the job.
- *Idle Time on a Machine* This is the time during which a machine does not have a job to process, i.e. idle time from the end of job (i – 1) to the start of job i.
- *Total Elapsed Time* This is the time interval between starting the first job and completing the last job, including the idle time (if any), in a particular order by the given set of machines.
- *No Passing Rule* This refers to the rule of maintaining the order in which jobs are to be processed on given machines. For example, if n jobs are to be processed on two machines, M_1 and M_2 in the order M_1M_2, then each job should go first to machine M_1 and then to M_2.

Assumptions:

1. No machine can process more than one operation at a time.
2. Each operation once started must be performed till completion.
3. Each operation must be completed before starting any other operation.
4. Time intervals for processing are independent of the order in which operations are performed.

5. There is only one machine of each type.
6. A job is processed as soon as possible, subject to the ordering requirements.
7. All jobs are known and are ready for processing, before the period under consideration begins.
8. The time required to transfer the jobs between machines is negligible.

Processing *n* Jobs Through Two Machines:

- Processing *n* jobs through two machines is a classic sequencing problem in operations research, often addressed using Johnson's Rule for optimal job sequencing. This problem falls under the category of two-machine flow shop scheduling and is typically framed to minimize the makespan, or total time required to complete all jobs.

Let there be *n* jobs, each of which is to be processed through two machines, M_1 and M_2 in the order $M_1 M_2$, i.e. each job has to pass through the same sequence of operations. In other words, a job is assigned on machine M_1 first and once processing is over on machine M_1, it is assigned to machine M_2. If the machine M_2 is not free for processing the same job, then the job is placed in waiting line for its turn on machine M_2, i.e. passing is not allowed.

Since passing is not allowed, therefore, machine M_1 will remain busy in processing all the *n* jobs one by – one, while machine M_2 may remain idle waiting for the jobs to come from M_1. The idle time for both M_1 and M_2 may be reduced by determining an optimal sequence of n jobs to be processed on two machines M_1 and M_2. The procedure suggested by ***Johnson*** for determining the optimal sequence is summarized as follows:

Johnson Procedure:

Step 1: List the jobs along with their processing times in a table as shown below:

Processing Time on Machine	Job Number *1*	*2*	*3*	…	*n*
M_1	t_{11}	t_{12}	t_{13}	…	t_{1n}
M_2	t_{21}	t_{21}	t_{21}	…	t_{2n}

Step 2: Examine the processing times on machines M_1 and M_2, in each column of the table and find the shortest processing time, i.e. find out, min. (t_{1j}, t_{2j}) for all $_j$.

Step 3(a): If the shortest processing time is on machine M_1, then place the job in the first available position in the sequence. If the processing time is on machine M_2, then place the job in the last available position in the sequence.

(b) If there is a tie in selecting the minimum of all the processing times, then the following three situations may arise:

i. If minimum processing time is same on both machines, i.e. min (t_{1j}, t_{2j}) = t_{1k} = t_{2r}, then process the kth job first and the rth job last.
ii. If minimum processing times, t_{1j} on machine M_1 are same for more than two jobs, then select the job corresponding to the smallest job subscript, j first.
iii. If minimum processing times, t_{2j} on machine M_2 are same for more than two jobs, then select the job corresponding to the largest job subscript, j last.

Step 4: Cross off assigned jobs from the table. If no job remains to be assigned, then stop the procedure and go to Step 5. Otherwise, go to Step 2.

Step 5: Calculate the idle time for machines M_1 and M_2:

a. Idle time for machine M_1
= (Total elapsed time) – (Time when the last job in a sequence finishes on machine M_1)

b. Idle time for machine M_2

$$\text{Time when first job in a sequence finishes on machine } M_1$$
$$+ \sum_{j=2}^{n} \{(\text{Time when the } j\text{th job in a sequence starts on machine } M_2)$$
$$= -(\text{Time when the } (j-1)\text{th job in a sequence finishes on machine } M_2)\}.$$

Step 6: The total elapsed time to process all jobs through two machines is given by:

Total elapsed time = Time when the nth job in a sequence finishes on machine M_2.

$$= \sum_{j=1}^{n} M_{2j} + \sum_{j=1}^{n} I_{2j}$$

where M_{2j} = Time required for processing jth job on machine M_2.

I_{2j} = Idle time for machine M_2 after processing (j – 1)th job and before the start of jth job processing.

Problems:

1. There are five jobs, each of which must go through the two machines *A* and *B* in the order *AB*. Processing times are given below.

Job	1	2	3	4	5
Machine A	5	1	9	3	10
Machine B	2	6	7	8	4

Determine a sequence for the five jobs that will minimize the total elapsed time.

Solution:

The shortest processing time in the given problem is 1 on machine A. So perform job 2 in the beginning, as shown below.

2				

The reduced list of processing time becomes

Job	1	3	4	5
Machine A	5	9	3	10
Machine B	2	7	8	4

Again the shortest processing time in the reduced list is 2 for job 1 on machine *B*. So place job 1 as the last.

2				1

Continuing in the same manner the next reduced list is obtained as

Job	3	4	5
Machine A	9	3	10
Machine B	7	8	4

Leading to the sequence

2	4			1

and the list

Job	3	5
Machine A	9	10
Machine B	7	4

gives rise to the sequence

2	4		5	1

Finally, the optimal sequence *n* is obtained as

2	4	3	5	1

Computation of the total elapsed time and the machine's idle time in hours.

Job	*Machine A*		*Machine B*		*Idle Time*	
	Time In	*Time Out*	*Time In*	*Time Out*	*A*	*B*
2	0	0+1=1	1	1+6=7	0	1
4	1	1+3=4	7	7+8=15	0	0
3	4	4+9=13	15	15+7=22	0	0
5	13	13+10=23	23	23+4=27	0	1
1	23	23+5=**28**	28	28+2=**30**	30-28=2	28-27=1
					2	1+1+1=3

From the above table we find that the total elapsed time is 30 hours and the idle time on Machine A is 2 hours and on Machine B is 3 hours.

2. Find the sequence that minimizes the total elapsed time (in hours) required to complete the following tasks on two machines.

Task	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*
Machine I	2	5	4	9	6	8	7	5	4
Machine II	6	8	7	4	3	9	3	8	11

Solution:

The shortest processing time is 2 hours on machine I for job *A*. Hence, process this job first.

A								

Deleting these jobs, we get the reduced list of processing time.

Task	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*
Machine I	5	4	9	6	8	7	5	4
Machine II	8	7	4	3	9	3	8	11

The next minimum processing time is same for jobs E and G on machine II. The corresponding processing time on machine I for this job is 6 and 7. The longest processing time is 7 hours. So sequence job G at the end and E next to it.

A							E	G

Deleting the jobs that are sequenced, the reduced processing list is,

Task	*B*	*C*	*D*	*F*	*H*	*I*
Machine I	5	4	9	8	5	4
Machine II	8	7	4	9	8	11

The minimum processing time is 4 hours for job C, I and D. For job C and I it is on machine I and for job D it is on machine II. There is a tie in sequencing jobs C and I. To break this, we consider the corresponding time on machine II, the longest time is 11 (eleven) hours. Hence, sequence job I in the beginning followed by job C. For job D, as it is on machine II, sequences it last.

A	I	C				D	E	G

Deleting the jobs that are sequenced, the reduced processing list is,

Task	*B*	*F*	*H*
Machine I	5	8	5
Machine II	8	9	8

The next minimum processing time is 5 hours on machine I for jobs B and H, which is again a tie. To break this, we consider the corresponding longest time on the other machine (II) and sequence job B or H first. Finally, job F is sequenced.

The optimal sequence for this job is,

A	I	C	B	H	F	D	E	G

The total elapsed time and idle time for both the machines are calculated from the following table.

Job	*Machine I*		*Machine II*		*Idle Time*	
	Time In	*Time Out*	*Time In*	*Time Out*	*I*	*II*
A	0	0+2=2	2	2+6=8	0	2
I	2	2+4=6	8	8+11=19	0	0
C	6	6+4=10	19	19+7=26	0	0
B	10	10+5=15	26	26+8=34	0	0
H	15	15+5=20	34	34+8=42	0	0
F	20	20+8=28	42	42+9=51	0	0
D	28	28+9=37	51	51+4=55	0	0
E	37	37+6=43	55	55+3=58	0	0
G	43	43+7=**50**	58	58+3=**61**	61-50=11	0
					11	2

Total elapsed time = 61 hours.
Idle time for *Machine I* = 11 hours; Idle time for *Machine II* = 2 hours.

3. A machine operator has to perform two operations, turning and threading, on a number of different jobs. The time required to perform these operations in minutes for each job is given. Determine the order in which the jobs should be processed in order to minimize the total time required to turn out all the jobs.

Jobs:	1	2	3	4	5	6
Time for turning (in min.)	3	12	5	2	9	11
Time for threading (in min).	8	10	9	6	3	1

Solution:
The smallest element is 1 in the given matrix and falls under second operation. Hence do the 6 th job last. Next smallest element is 2 for the job 4 and falls under first operation hence do the fourth job first. Next smallest element is 3 for job 1 falls under first operation hence do the first job second. Like this go on proceed until all jobs are over. The optimal sequence is:

4	1	3	2	5	6

Total Elapsed Time and Idle Time is given the table:

Optimal Sequence	*Turning Operation*		*Threading Operation*		*Machine Idle.*		*Job idle*
	In	*Out*	*In*	*Out*	*Turning*	*Threading*	
4	0	0+2=2	2	2+6=8	0	2	0
1	2	2+3=5	8	8+8=16	0	0	3
3	5	5+5=10	16	16+9=25	0	0	6
2	10	10+12=22	25	25+10=35	0	0	3
5	22	22+9=31	35	35+3=38	0	0	4
6	31	31+11=**42**	42	42+1=**42**	42-42=0	0	0
					0	2	

4. A book binder has one printing press, one binding machine and manuscripts of 7 different books. The times required for performing printing and binding operations for different books are shown below:

Book	1	2	3	4	5	6	7
Printing time (hours):	20	90	80	20	120	15	65
Binding time (hours):	25	60	75	30	90	35	50

Decide the optimum sequence of processing of books in order to minimize the total time required to bring out all the books.

Solution: Examine both printing and binding operations time column wise. The smallest value is 15 printing hours for book 6. The book 6 is placed first in the sequence and cross off column 6 of the table. Repeat the procedure of finding smallest processing time column wise. Books 1 and 4 are placed in the sequence as shown below:

6	1	4				

Cross off columns 1, 4 and 6 of the table. For the remaining columns 2, 3, 5 and 7 the smallest time is 50 binding hours for book 7, so it should be sequenced in the last as shown below:

6	1	4				7

After this sequence, remaining set of printing and binding times gets reduced as follows:

Book	2	3	5
Printing time (hours):	90	80	120
Binding time (hours):	60	75	90

The smallest time in this reduced data is 60, which corresponds to binding. So book 2 is placed in the second cell from right. Proceeding in the same manner, the optimal sequence so obtained is shown below:

6	1	4	5	3	2	7

The total minimum elapsed time for printing and binding is given in Table

Book	Printing		Binding		Idle Time
	Time In	Time Out	Time In	Time Out	
6	0	0+15=15	15	15 + 35 = 50	15
1	15	15+20=35	50	50 + 25 = 75	—
4	35	35+20=55	75	75 + 30 = 105	—
5	55	55+120=175	175	175 + 90 = 265	70
3	175	175+80=255	265	265 + 75 = 340	—
2	255	255+90=345	345	345 + 60 = 405	5
7	345	345+65=410	410	410 + 50 = 460	5

The minimum elapsed time, i.e. time from start of printing book 6 to binding the last book 7 is 460 hours. During this time the printing machine remains idle for 460 – 410 = 50 hours.

5. A manufacturing company processes 6 different jobs on two machines A and B. Number of units of each job and its processing times on A and B are given in the following table. Find the optimum sequence, the total minimum elapsed time and idle time for each machine.

Job Number	No of Units on Each Job	Processing Time (Hours)	
		Machine A	Machine B
1	3	5	8
2	4	16	7
3	2	6	11
4	5	3	5
5	2	9	7.5
6	3	6	14

Solution:

Examine processing time of machines A and B. The smallest value is 3 minutes for job 4 on machine A. Thus job 4 is scheduled first in the sequence as shown below:

4					

Repeating this procedure, the optimal sequence so obtained is shown below.

4	1	3	6	5	2
5	3	2	3	2	4

No of Units on Each Job

The total minimum elapsed time and idle time for each machine:

Job Number	Unit Number of Job	Machine A		Machine B		Idle Times of Machine (min)
		Time In	Time Out	Time In	Time Out	
4	1	0	0+3=3	3	3+5=8	3
	2	3	3+3=6	8	8+5=13	0
	3	6	6+3=9	18	13+5=18	0
	4	9	9+3=12	23	18+5=23	0
	5	12	12+3=15	28	23+5=28	0
1	1	15	15+5=20	28	28+8=36	0
	2	20	20+5=25	36	36+8=44	0
	3	25	25+5=30	44	44+8=52	0
3	1	30	30+6=36	52	52+11=63	0
	2	36	36+6=42	63	63+11=74	0
6	1	42	42+6=48	74	74+14=88	0
	2	48	48+6=54	88	88+14=102	0
	3	54	54+6=60	102	102+14=116	0
5	1	60	60+9=69	116	116+7.5=123.5	0
	2	69	69+9=78	123.5	123.5+7.5=131	0
2	1	78	78+16=94	131	131+7=138	0
	2	94	94+16=110	138	138+7=145	0
	3	110	110+16=126	145	145+7=152	0
	4	126	126+16=142	152	152+7=159	0

The total elapsed time for all the jobs including the number of units is 159 minutes. However, machine A remained idle for 17 (= 159 – 142) minutes and machine B remained idle for 3 (3 + 152 – 152) minutes.

6. There are seven jobs, each of which has to go through the machines A and B in the order AB. Processing times in hours are as follows:

Job	1	2	3	4	5	6	7
Machine A	3	12	15	6	10	11	9
Machine B	8	10	10	6	12	1	3

Determine a sequence of these jobs that will minimize the total elapsed time T. Also find T and idle time for machines A and B.

Solution:

The smallest processing time is 1 hour for job 6 on machine B. Thus job 6 will be processed last on machine A as shown below:

						6

The reduced set of processing times becomes

Job	1	2	3	4	5	7
Machine A	3	12	15	6	10	9
Machine B	8	10	10	6	12	3

There are two equal minimal values: processing time of 3 hours for job 1 on machine A and processing time of 3 hours for job 7 on machine B. According to Johnson's rules, job 1 is scheduled first and job 7 next to 6 as shown below:

1					7	6

The reduced set of processing times becomes

Job	2	3	4	5
Machine A	12	15	6	10
Machine B	10	10	6	12

Again there are two equal minimal values: processing time of 6 hours for job 4 on machine A as well as on machine B. We may choose arbitrarily to process (schedule) job 4 next to job 1 or next to job 7 as shown below:

1	4				7	6

or

1				4	7	6

The reduced set of processing times becomes

Job	2	3	5
Machine A	12	15	10
Machine B	10	10	12

There are three equal minimal values: processing time of 10 hours for job 5 on machine A and for jobs 2 and 3 on machine B. According to rules: job 5 is scheduled next to job 4 in the first schedule or next to job 1 in the second schedule. Job 2 then is scheduled next to job 7 in the first schedule or next to job 4 in the second schedule. The optimal sequences are shown below:

1	4	5	3	2	7	6

or

1	5	3	2	4	7	6

The calculations for total elapsed time and idle times for machines A and B are shown in Table

Job	*Machine A*		*Machine B*		*Idle Time for Machine B*
	Time In	*Time Out*	*Time In*	*Time Out*	
1	0	0+3=3	3	3+8=11	3
4	3	3+6=9	11	11+6=17	0
5	9	9+10=19	19	19+12=31	2
3	19	19+15=34	34	34+10=44	3
2	34	34+12=46	46	46+10=56	2
7	46	46+9=55	56	56+3=59	0
6	55	55+11=66	66	66+1=67	7
					17

The minimum elapsed time is 67 hours. Idle time for machine A is 1 hours (66th–67th hour) and for Machine B is 17 hours.

Practice Problems:

1. We have five jobs, each of which must be processed on the two machines A and B, in the order AB. Processing times in hours are given in the table below:

Job	1	2	3	4	5
Machine A	5	1	9	3	10
Machine B	2	6	7	8	4

   Determine a sequence for the five jobs that will minimize the elapsed time T.

   (Ans: Optimal sequence: 2 – 4 – 3 – 5 – 1; Elapsed time = 30 hours. Idle time for machine A = 2 hours, and for machine B = 3 hours.)

2. Find the sequence that minimizes the total elapsed time to complete the following jobs. Each job is processed in the order (i) AB (ii)BA

Job	1	2	3	4	5	6	7
Machine A (min)	12	6	5	11	5	7	6
Machine B (min)	7	8	9	4	7	8	3

   Determine the sequence for the jobs so as to minimize the process time. Also find total elapsed time.

   (Ans. (i) 3 – 5 – 2 – 6 – 1 – 1 – 7; 51 min; 3 min (ii) 7 – 4 – 1 – 6 – 2 – 5 – 3; 55 min; 9 min)

3. A book binder has one printing press, one binding machine, and manuscripts of a number of books. The time required to perform the printing and binding operations on each book are shown below. The binder wishes to determine the order in which the books should be processed, so that the total time required to process all books is minimized.

Book	1	2	3	4	5	6
Printing Time (hrs)	30	120	50	20	90	110
Binding Time (hrs)	80	100	90	60	30	10

(Ans: Optimal sequence: 4 – 1 – 3 – 2 – 5 – 6; Elapsed time = 430 hours. Idle time for printing machine = 10 hours, and for binding machine = 40 hours.)

4. A readymade garment manufacturer has to process 7 items through two stages of production, viz. cutting and sewing. The time taken for each of these items at the different stages is given below in appropriate units.

Item	1	2	3	4	5	6	7
Cutting Time	5	7	3	4	6	7	12
Sewing Time	2	6	7	5	9	5	8

Find the order in which these items are to be processed through these stages so as to minimize the total processing time.

(Ans: 3 – 4 – 5 – 7 – 2 6 – 1)

5. Five jobs are performed, first on machine X and then on machine Y. The time taken, in hours by each job on each machine is given below:

Job	A	B	C	D	E
Time on machine X	12	4	20	14	22
Time on machine Y	3	14	16	18	10

Determine the optimum sequence of jobs that minimizes the total elapsed time to complete the jobs. Also compute the minimum time.

(Ans: Optimal sequence: B – D – C – E – A; Elapsed time = 48 hours. Idle time is 12 hours for machines X and Y each.)

6. The following table shows the machine time (in hours) for 5 jobs to be processed on two different machines:

Job	1	2	3	4	5
Machine A (min)	3	7	4	5	7
Machine B (min)	6	2	7	3	4

Passing is not allowed. Find the optimal sequence in which jobs should be processed.

(Ans: Optimal sequence: 1 – 3 – 5 – 4 – 2; Elapsed time = 28 hours. Idle time for machine A = 2 hours, and for machine B = 3 hours.)

7. Six jobs go over machine I first and then over II. The order of the completion of jobs has no significance. The following table gives the machine times in hours for six jobs and the two machines.

Job	1	2	3	4	5	6
Machine I	5	9	4	7	8	6
Machine B II	7	4	8	3	9	5

Find the sequence of the jobs that minimizes the total elapsed time for completing the jobs. Find the minimum time by using Gantt Chart or by any other method.

(Ans: Optimal sequence: 3 – 1 – 5 – 6 – 2 – 4; Elapsed time = 35 hours.)

8. We have five jobs, each of which must go through two machines in the order AB. Their processing times are given below:

Job	1	2	3	4	5
Machine A	10	2	18	6	20
Machine B	4	12	14	16	18

(Ans: Optimal sequence: 2 – 4 – 3 – 5 –1; Elapsed time = 60 hours.)

Processing *n* Jobs Through Three Machines:

Consider n jobs (1, 2 ... n) processing on three machines M_1, M_2 and M_3. The list of jobs with their processing times on three machines M_1, M_2 and M_3 is given below. An optimal solution to this problem can be obtained if either or both of the following conditions hold good:

Processing Time on Machine	Job Number *1*	*2*	*3*	…	*n*
M_1	t_{11}	t_{12}	t_{13}	…	t_{1n}
M_2	t_{21}	t_{21}	t_{21}	…	t_{2n}
M_3	t_{31}	t_{31}	t_{31}	…	t_{3n}

1. The minimum processing time on machine M_1 is at least as great as the maximum processing time on machine M_2, that is, min $t_{1j} \geq$ max t_{2j}, for j = 1, 2, . . ., n
2. The minimum processing time on machine M_3 is at least as great as the maximum processing time on machine M_2, that is, min $t_{3j} \geq$ max t2j, for j = 1, 2, . . ., n

If either or both the above conditions hold good, then the steps of the algorithm can be summarized in the following steps:

Procedure:

Step 1: Examine the processing times of the given jobs on all three machines and if either one or both the above conditions hold, then go to Step 2, otherwise the algorithm fails.

Step 2: Introduce two fictitious machines, say G and H with corresponding processing times given by:

i. $t_{Gj} = t_{1j} + t_{2j}$, j = 1, 2, . . ., n
that is, the processing time on machine G is the sum of the processing times on machines M_1 and M_2, and

ii. $t_{Hj} = t_{2j} + t_{3j}$, j = 1, 2, . . ., n
that is, processing time on machine H is the sum of the processing times on machines M_2 and M_3.

Step 3: Determine the optimal sequence of jobs for this n – job, two machine equivalent sequencing problem with the prescribed ordering GH in the same way as discussed earlier.

Problems:

1. Find the sequence that minimizes the total time required in performing the following jobs on three machines in the order ABC. Processing times (in hours) are given in the following table:

Job	1	2	3	4	5
Machine A	8	10	6	7	11
Machine B	5	6	2	3	4
Machine C	4	9	8	6	5

Solution:

From data of the problem, we know that min (t_{Aj}) = 6; min (t_{Cj}) = 4; max (t_{Bj}) = 6. Since min (tAj) ≥ (t_{Bj}) for all j is satisfied, the given problem can be converted into a problem of 5 jobs and two machines. The processing time on two dummy machines G and H can be determined by the following relationships:

$t_{Gj} = t_{Aj} + t_{Bj}$; and $t_{Hj} = t_{Bj} + t_{Cj}$; j = 1, 2, . . ., n

The processing times for the new problem are given below:

Job	1	2	3	4	5
Machine G	8 + 5 = 13	10 + 6 = 16	6 + 2 = 8	7 + 3 = 10	11 + 4 = 15
Machine H	5 + 4=9	6 + 9 = 15	2 + 8 = 10	3 + 6 = 9	4 + 5 = 9

When the procedure described for n jobs on two machines is applied to this problem, the optimal sequence, so obtained, is given by:

3	2	5	1	4

The total minimum elapsed time is given in Table

Job Sequence	Machine A		Machine B		Machine C	
	Time In	Time Out	Time In	Time Out	Time In	Time Out
3	0	0+6=6	6	6+2=8	8	8+8=16
2	6	6+10=16	16	16+6=22	22	22+9=31
5	16	16+11=27	27	27+4=31	31	31+5=36
1	27	27+8=35	35	35+5=40	40	40+4=44
4	35	35+7=42	42	42+3=45	45	45+6=51

Table indicates that the minimum total elapsed time is 51 hours. The idle time for machines A, B and C is 9 (= 51 – 42) hours, 6 (= 51 – 45) hours and 9 {= (8 – 0) + (45 – 44)} hours, respectively.

2. A machine operator has to perform three operations, turning, threading and knurling, on a number of different jobs. The time required to perform these operations (in minutes) on each job is known. Determine the order in which the jobs should be processed in order to minimize the total time required to turn out all the jobs. Also find the minimum elapsed time.

Job	1	2	3	4	5	6
Turning	3	12	5	2	9	11
Threading	8	6	4	6	3	1
Knurling	13	14	9	12	8	13

Solution:

Let us consider the three machines as A, B and C.

A = Turning; B = Threading; C = Knurling

i. Min (Ai Ci) = (2, 8)
   Max (Bi) = 8

ii. Min Ai = 2 Not greater than Max Bi = 8
   Min Ci = 8 ≥ Max Bi is satisfied.

Define two machines G and H

Such that, Gi = Ai + Bi

Hi = Bi + Ci

Job	1	2	3	4	5	6
Machine G	3 + 8 = 11	12 + 6 = 18	5 + 4 = 9	2 + 6 = 8	9 + 3 = 12	11 + 1 = 12
Machine H	8 + 13 =21	6 + 14 = 20	4 + 9 = 13	6 + 12 = 18	3 + 8 = 11	1 + 13 = 14

We adopt Johnson's algorithm steps to get the optimum sequence.

4	3	1	6	2	5

To find the min. total elapsed time and idle time for machines A, B and C

Job Sequence	Machine A		Machine B		Machine C	
	Time In	Time Out	Time In	Time Out	Time In	Time Out
4	0	0+2=2	2	2+6=8	8	8+12=20
3	2	2+5=7	8	8+4=12	20	20+9=29
1	7	7+3=10	12	12+8=20	29	29+13=42
6	10	10+11=21	21	21+1=22	42	42+13=55
2	21	21+12=33	33	33+6=39	55	55+14=69
5	33	33+9=42	42	42+3=45	69	69+8=77

Table indicates that the minimum total elapsed time is 77 hours. The idle time for machines A, B and C is 35 (= 77 – 42) hours, 47 {= (21 – 20) + (33 – 22) + (42 – 39) + (77 – 45)} hours and 8 {= (8 – 0) + (69 – 69)} hours, respectively.

3. A readymade garments manufacturer has to process 7 items through two stages of production, viz., cutting and sewing. The time taken for each of these items at the different stages are given below in appropriate units:

Item	1	2	3	4	5	6	7
Cutting Process Time	5	7	3	4	6	7	12
Sewing Process Time	2	6	7	5	9	5	8

i. Find an order in which these items are to be processed through these stages so as to minimize the total processing time.

ii. Suppose a third stage of production is added, viz. pressing and packing, with processing time for these items as follows:

Item	1	2	3	4	5	6	7
Printing and Packing Process Time	10	12	11	13	12	10	11

Find an order in which these seven items are to be processed so as to minimize the time taken to process all the items through all the three stages.

Solution:

Using Johnson's optimal sequencing procedure, the optimal sequence so obtained is shown below. The calculations for total elapsed time is shown in Table

3	4	5	7	2	6	1

Total elapsed time and idle time is shown in the table below:

Item	Cutting		Sewing		Idle time for Sewing
	Time In	Time Out	Time In	Time Out	
3	0	0+3=3	3	3+7=10	3
4	3	3+4=7	10	10+5=15	0
5	7	7+6=13	15	15+9=24	0
7	13	13+12=25	25	25+8=33	1
6	25	25+7=32	33	33+5=38	0
1	32	32+5=37	38	38+2=40	0

The total elapsed time is 46 hours. Idle item on cutting machine is 2 (= 46 – 44) hours and on sewing machine 4 hours.

ii. When the third stage of production (viz., pressing and packing) is added, the problem is that of seven jobs on three machines with processing time (in hours) given below:

Item	1	2	3	4	5	6	7
Cutting, *Ci*	5	7	3	4	6	7	12
Sewing, *Si*	2	6	7	5	9	5	8
Pressing, *Pi*	10	12	11	13	12	10	11

Given that min (Ci) = 3, min (Pi) = 10 and max (Si) = 9. Since the condition min (Pi) ≥ max (Si) for all i is satisfied, the problem can be converted into that of 7 jobs and 2 machines. If G and H are two fictitious machines such that Gi = Ci + Si and Hi = Si + Pi for all i, then the given problem can be re-written as the problem of two machines and 7 items with their processing times given below:

Item	1	2	3	4	5	6	7
Gi	5 + 2 = 7	7 + 6 = 13	3 + 7 = 11	4 + 5 = 9	6 + 9 = 15	7 + 5 = 12	12 + 8 = 20
Hi	2 + 10 =12	6 + 12 = 18	7 + 11 = 18	5 + 13 = 18	9 + 12 = 21	5 + 10 = 15	8 + 11 = 19

Using the optimal sequence algorithm, the optimal sequence so obtained is as follows:

1	4	3	6	2	5	7

The total elapsed time for three processes is given in Table

Item	*Cutting*		*Sewing*		*Printing and Packing*	
	Time In	Time Out	Time In	Time Out	Time In	Time Out
1	0	0+5=5	5	5+2=7	7	7+10=17

4	5	5+4=9	9	9+5=14	17	17+13=30
3	9	9+3=12	14	14+7=21	30	30+11=41
6	12	12+7=21	21	21+5=26	41	41+10=51
2	21	21+7=27	28	28+6=34	51	51+12=63
5	27	27+6=33	34	34+9=43	63	63+12=75
7	33	33+12=45	45	45+8=53	75	75+11=86

The minimum total elapsed time is 86 hours with idle time of 42 hours for cutting, 44 hours for sewing and 7 hours for pressing and packing.

4. We have five jobs, each of which must go through the machines A, B and C in the order ABC. Determine the sequence that will minimize the total elapsed time.

Job No	*1*	*2*	*3*	*4*	*5*
Machine A	5	7	6	9	5
Machine B	2	1	4	5	3
Machine C	3	7	5	6	7

Solution:

The optimum sequence can be obtained by converting the problem into that of two-machines, by using the following steps.

Step 1 Find Min (Ai Ci) i = 1, 2 ... 5

= (5, 3).

Step 2 Max (Bi) = 5

Min Ai = 5 = Max Bi = 5

∴ Min Ai ≥ Max Bi is satisfied.

We convert the problem into a two-machine problem by defining two machines G and H, such that the processing time on G and H are given by,

Gi = Ai + Bi

Hi = Bi + Ci

Job	*1*	*2*	*3*	*4*	*5*
G	7	8	10	14	8
H	5	8	9	11	10

We obtain the optimum sequence by using the steps in Johnson's algorithm.

5	2	4	3	1

To find the total elapsed time and idle time on three machines.

Job	*Machine A*		*Machine B*		*Machine C*		*Idle Time*		
	Time In	Time Out	Time In	Time Out	Time In	Time Out	*A*	*B*	*C*
5	0	0+5=5	5	5+3=8	8	8+7=15	0	5	8
2	5	5+7=12	12	12+1=13	15	15+7=22	0	4	0
4	12	12+9=21	21	21+5=26	26	26+6=32	0	8	4
3	21	21+6=27	27	27+4=31	32	32+5=37	0	1	0
1	27	27+5=32	32	32+2=34	37	37+3=40	0	1	0

Total elapsed time = 40 hours.

Idle time for machine A = 8 (40 – 32 = 8) hours;

Idle time for machine B = 25 (40 – 34 + 8+4+8+1+1 = 25) hours;

Idle time for machine C = 12 hours.

Practice Problems:

1. A machine operator has to perform three operations, turning, threading and knurling on a number of different jobs. The time required to perform these operations (in minutes) for each job is known and is give below:

Job	Time (min)		
	Turning	Threading	Knurling

1	3	8	13
2	12	6	14
3	5	4	9
4	2	6	12
5	9	3	8
6	11	1	13

Determine the order in which the jobs should be processed in order to minimize the total time required to turn out all the jobs.

(Ans: Optimal sequences: (i) 4 – 3 – 1 – 6 – 2 – 5; Elapsed time = 77 minutes; Idle time is 35 minutes for turning operation; 49 minutes for threading operation and 8 minutes for knurling operation.)

2. A foreman wants to process four different jobs on three machines: a shaping machine, a drilling machine and a tapping machine, the sequence of operations being shaping – drilling – tapping. Decide the optimal sequence for the four jobs to minimize the time elapsed from the start of first job to the end of last job if the process times are

Job	Shaping	Drilling	Tapping
1	13	3	18
2	18	8	4
3	8	6	13
4	23	6	8

(Ans: 3 – 1 – 4 – 2; Elapsed time = 74 minutes; Idle time is 12, 51 and 31 minutes respectively)

3. Determine the optimal sequence of jobs that minimize the total elapsed time, based on the following information. Processing time on machines is given in hours and passing is not allowed.

Job No	*A*	*B*	*C*	*D*	*E*	*F*	*G*
Machine M_1	3	8	7	4	9	8	7
Machine M_2	4	3	2	5	1	4	3
Machine M_3	6	7	5	11	5	6	12

(Ans: Optimal sequences: D – G – F – B – C – E; Elapsed time = 59 hours. Idle time is 13 hours for machine M_1; 37 hours for machine M_2 and 7 hours for machine M_3.)

4. Find the sequence, for the following eight jobs, that will minimize the total elapsed time for the completion of all the jobs. Each job is processed in the same order CAB. Entries give the time in hours on the machines.

Job No		*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*
Times on machines	*A*	4	6	7	4	5	3	6	2
	B	8	10	7	8	11	8	9	13
	C	5	6	2	3	4	9	15	11

(Ans: 4 – 1 – 3 – 5 – 2 – 8 – 7 - 6; Tmin = 81 hours.)

5. We have six jobs, each of which must go through machines A, B and C in the order ABC. Processing time (in hours) are given in the following table:

Job No	*1*	*2*	*3*	*4*	*5*	*6*
Machine A	8	3	7	2	5	1
Machine B	3	4	5	2	1	6
Machine C	8	7	6	9	10	9

Determine a sequence for the five jobs that will minimize the elapsed time t.

(Ans: Optimal sequences: (i) 4 – 5 – 2 – 6 – 1 – 3, (ii) 4 – 5 – 2 – 6 – 1 – 3; Elapsed time = 53 hours. Idle time is 27 hours for machine A; 32 hours for machine B and 4 hours for machine C.)

Processing *n* Jobs Through *m* Machines:

Let there be *n* jobs, each of which is to be processed through m machines, say $M_1, M_2, \ldots, M_m$ in the order $M_1\ M_2 \ldots M_m$. The optimal solution to this problem can be obtained if either or both of the following conditions hold good.

i. Min $\{t_{1j}\} \geq$ Max $\{t_{ij}\}$; $j = 2, 3, \ldots, m-1$ and/or

ii. Min $\{t_{mj}\} \geq$ Max $\{t_{ij}\}$; $j = 2, 3, \ldots, m-1$

that is, the minimum processing time on machines M_1 and M_m is as great as the maximum processing time on any of the remaining (m – 1) machines.

If either or both these conditions hold good, then the steps of the algorithm can be summarized in the following steps:

Step 1: Find, Min {t1j}, Min {tmj} and Max {tij} and verify the above conditions. If either or both the conditions mentioned above hold, then go to Step 2. Otherwise the algorithm fails.

Step 2: Convert m-machine problem into 2-machine problem by introducing two fictitious machines, say G and H with corresponding processing times given by:

i. $tGj = t1j + t2j + \ldots + tm-1j$; $j = 1, 2, \ldots, n$

i.e. processing time of n-jobs on machine G is the sum of the processing times on machines $M_1, M_2, \ldots, M_{m-1}$, j

ii. $tHj = t2j + t3j + \ldots + tmj$; $j = 1, 2, \ldots, n$

i.e. processing time of n-jobs on machine H is the sum of the processing times on machines $M_2, M_3, \ldots, M_m$.

Step 3: The new processing times, so obtained, can now be used for solving n-job, two-machine equivalent sequencing problem with the prescribed ordering HG in the same way as discussed earlier.

Remarks

1. In addition to the conditions given in Step 2, if:
   $t2j + t3j + \ldots + tm - 1, j = k$ (constant)

for all $j = 1, 2, \ldots, m-1$, then the optimal sequence can be obtained for n-jobs and two machines M_1 and M_m in the order M_1M_m as usual.

2. If $t_{1j} = t_{mj}$ and $tGj = tHj$, for all $j = 1, 2, \ldots, n$, then the total number of optimal sequences will be n and total minimum elapsed time in these cases would also be the same.
3. The method described above for solving n-jobs and m-machines sequencing problem is not a general method. It is applicable only to certain problems where the minimum cost (or time) of processing the jobs through first and/or last machine is more than or equal to the cost (or time) of processing the jobs through the remaining machines.

Problems:

1. Four jobs 1, 2, 3 and 4 are to be processed on each of the five machines A, B, C, D and E in the order A B C D E. Find the total minimum elapsed time if no passing of jobs is permitted. Also find the idle time for each machine.

Machines	*Jobs*			
	1	*2*	*3*	*4*
A	7	6	5	8
B	5	6	4	3
C	2	4	5	3
D	3	5	6	2
E	9	10	8	6

Solution:

Since the problem is to be sequenced on five machines, we convert the problem into a two – machine problem by adopting the following steps.

Step 1: Find Min (Ai, Ei) = (5, 6) i = 1, 2, 3, 4

Max (Bi, Ci, Di) = (6, 5, 6)

Step 2: The inequality

Min Ei = 6 ≥ Max (Bi, Ci, Di) is satisfied.

Therefore, we can convert the problem into a two-machine problem.

Step 3: Since Bi + Ci + Di ≠ C, where C is a fixed constant, we define two machines G and H such that,

Gi = Ai + Bi + Ci + Di

Hi = Bi + Ci + Di + Ei i = 1, 2, 3, 4.

Job	*1*	*2*	*3*	*4*
G	17	21	20	16
H	19	25	23	24

We obtain the optimum sequence by using the steps in Johnson's algorithm.

1	3	2	4

Total elapsed time and idle time on five machines.

Job	*Machine A*		*Machine B*		*Machine C*		*Machine D*		*Machine E*		*Idle Time*				
	In	Out	In	Out	In	Out	In	Out	In	Out	*A*	*B*	*C*	*D*	*E*
1	0	7	7	12	12	14	14	17	17	26	0	7	12	14	17
3	7	12	12	16	14	19	19	25	26	34	0	0	0	2	0
2	12	18	18	24	24	28	28	33	44	44	0	2	5	3	0
4	18	26	26	29	29	32	33	35	44	50	0	2	1	0	0

Total elapsed time = 50 hours.

Idle time for machine *A* = 24 hours; (50 – 26)

Idle time for machine *B* = 32 hours; (50 – 29) + 11

Idle time for machine *C* = 36 hours; (50 – 32) + 18

Idle time for machine *D* = 34 hours; (50 – 35) + 19

Idle time for machine *E* = 17 hours.

2. Find an optimal sequence for the following sequencing problems of four jobs and five machines, when passing is not allowed. Its processing time (in hours) is given below:

Job	Machines				
	M_1	M_2	M_3	M_4	M_5
A	7	5	2	3	9
B	6	6	4	5	10
C	5	4	5	6	8
D	8	3	3	2	6

Also find the total elapsed time.

Solution:

Here, Min (tM1 , j) = 5 = tM1 ,C ; Min (tM5 , j) = 6 = tM5 ,D

and Max {tM2 , j , tM3 , j , tM4 , j}= {6, 5, 6} respectively.

Since the condition of Min (tM5 , j) ≥ Max otM2 , j , tM3 , j , tM4 , jt is satisfied, therefore the given problem can be converted into a four jobs and two machines problem as G and H. The processing times of four jobs denoted by tGj and tHj on G and H, respectively are as follows:

Job	A	B	C	D
Machine G	17	21	20	16
Machine H	19	25	23	14

where $t_{Gj} = \sum_{i=1}^{m-1} t_{ij}$ and $t_{Hj} = \sum_{i=2}^{m} t_{ij}$.

Now, using the optimal sequence algorithm, the following optimal sequence can be obtained

A	C	B	D

The total elapsed time, corresponding to the optimal sequence can be calculated as shown in Table, using the individual processing times given in the original problem.

Job Sequence	Machines				
	M_1	M_2	M_3	M_4	M_5
A	0 – 7	7 – 12	12 – 14	14 – 17	12 – 26
C	7 – 12	12 – 16	16 – 21	21 – 27	27 – 35
B	12 – 18	18 – 24	24 – 28	28 – 33	35 – 45
D	18 – 26	26 – 29	29 – 32	33 – 35	45 – 51

Table shows that the minimum total elapsed time is 51 hours.
The idle time for machines M_1, M_2, M_3, M_4 and M_5 is 25, 33, 37 and 18 hrs, respectively.

3. Solve the following sequencing problem, giving an optimal solution when passing is not allowed.

Machine	Job				
	A	B	C	D	E
M_1	11	13	9	16	17
M_2	6	7	5	8	4
M_3	6	7	5	8	4
M_4	15	8	13	9	11

Solution:
From the data of the problem it is observed that

$$\text{Min}\,(t_{M_1,j}) = 9 = t_{M_1,C}\,; \quad \text{Min}\,(t_{M_4,j}) = 8 = t_{M_4,B}$$

and $$\text{Max}\,\{t_{M_2,j}\} = 6 = t_{M_2,E}\,; \quad \text{Max}\,\{t_{M_3,j}\} = 8 = t_{M_3,D}$$

Since both the conditions

$$\text{Min}\,(t_{M_1,j}) \geq \text{Max}\,\{t_{M_2,j}\,;\, t_{M_3,j}\}$$

$$\text{Min}\,(t_{M_4,j}) \geq \text{Max}\,\{t_{M_2,j}\,;\, t_{M_3,j}\}\,; \quad j = 1, 2, \ldots, 5$$

are satisfied, therefore the given problem can be converted into a 5-jobs and 2-machine problem as G and H. Further, it may be noted that, $t_{M2,j} + t_{M3,j} = 10$ (a fixed constant) for all j (j = 1, 2, . . ., 5). Thus, the given problem is reduced to a problem of solving 5 – jobs through 2 – machines M_1 and M_4 in the order M_1M_4. This means machines M_2 and M_4 will have no effect on the optimality of the sequences.
The processing times of 5 jobs on machine M_1 and M_4 is as follows:

Job	A	B	C	D	E
Machine M_1	11	13	9	16	17
Machine M_4	15	8	13	9	11

Now, using the algorithm described earlier, the optimal sequence so obtained is as follows:

C	A	E	D	B

The total elapsed time corresponding to the optimal sequence is 83 hours as shown in Table, using the individual processing times given in the original problem:

Job Sequence	Machines			
	M_1	M_2	M_3	M_4
C	0 – 9	9 – 14	14 – 19	19 – 32
A	9 – 20	20 – 24	24 – 30	32 – 45
E	29 – 36	36 – 42	42 – 46	46 – 57
D	36 – 52	52 – 54	54 – 62	62 – 71
B	52 – 65	65 – 68	68 – 75	75 – 83

4. Four jobs 1, 2, 3 and 4 are to be processed on each of the five machines A, B, C, D and E in the order ABCDE. Find the total minimum elapsed time if no passing of jobs is permitted. Also determine idle time for each machine.

Job	Machines				
	A	B	C	D	E
1	7	5	2	3	9
2	6	6	4	5	10
3	5	4	5	6	8
4	8	3	3	2	6

Solution:

Here min Ai = 8, min Ei = 6 and max (Bi, Ci, Di) = (6, 5 6). Since min Ei = max (Bi, Di), the equivalent problem involving 4 jobs 2 machines G and H becomes

Job	Machine G	Machine H
1	17	19
2	21	25
3	20	23
4	16	14

Examining the columns, we find that the optimal sequence is

1	3	2	4

Now calculate the total elapsed time corresponding to the optimal sequence , using the individual processing times.

Job Sequence	Machines				
	A	B	C	D	E
1	0 – 7	7 – 12	12 – 14	14 – 17	17 – 26
3	7 – 12	12 – 16	16 – 21	21 – 27	27 – 35
2	12 – 18	18 – 24	24 – 28	28 – 33	35 – 45
4	18 – 26	26 – 29	29 – 32	33 – 35	45 – 51

Thus the minimized elapsed time is 51 time units. Idle time on machine

A = 51 – 26 = 25 time units

B = 7 + 2 + 2 + (51 – 29) = 33 time units

C = 12 + 2 + 3 + 1 + (51 – 32) = 37 time units

D = 14 + 1 + + (51 – 35) = 35 time units

And E = 17 + 1 = 18 time units

5. Determine an optimal sequence for the folowingnsequencing problem:

Job	M_1	M_2	M_3	M_4
A	20	10	9	20
B	17	7	15	17
C	21	8	10	21
D	25	5	9	25

Solution:

Here min M_{i1} = 17, min M_{i4} = 17 and max M_{i2} = 10, M_{i3} = 15. Since min (M_{i1}, M_{i4}) > max (M_2, M_{i3}), thus Gi and Hi are two machines such that

$M_{iG} = M_{i1} + M_{i2} + M_{i3}$

$M_{iH} = M_{i2} + M_{i3} + M_{i4}$

Job	*Machine Gi*	*Machine Hi*
A	39	39
B	39	39
C	39	39
D	39	39

Here, Since Gi = Hi, i = 1, 2, 3, 4 and Mi1 = Mi4, there will be 4! = 24 schedules each giving us an optimal sequence. They are

ABCD	ABDC	ACBD	ACDB	ADBC	ADCB
BACD	BADC	BCAD	BCDA	BDAC	BDCA
CABD	CADB	CBAD	CBDA	CDAB	CSBA
DABC	DACB	DBAC	DBCA	DCAB	DCBA

Each of the above optimal sequence yields the same total elapsed time and given by table

Job	`Machines			
	M_1	M_2	M_3	M_4
A	0 – 20	20 – 30	30 – 39	39 – 59
B	20 – 37	37 – 44	44 – 59	59 – 76
C	37 – 58	58 – 66	66 – 76	76 – 97
D	58 – 83	83 – 88	88 – 97	97 – 122

Thus the total minimum elapsed time for each of the above 24 sequences is 122 hours.

Practice Problems:

1. There are four jobs each of which has to be processed on machines A, B, C, D, E and F in the order ABCDEF. Processing time in hours is given in the table below. Find the optimal sequencing of jobs, mimimum time required to process these jobs and the idle time for each of the machines.

Job	A	B	C	D	E	F
1	15	8	6	14	6	36
2	17	7	9	10	15	22
3	21	7	12	9	11	19
4	18	6	11	12	14	17

(Ans: 1 – 2 – 4 – 3; Tmin = 133 hrs; Idle times: 62, 105, 95, 88, 87 and 49 hrs respectively)

2. Find an optimal sequence for the following sequencing problem of four jobs and five machines (when passing is not allowed) of which processing time (in hrs) is as follows.

Job	1	2	3	4
Machine M_1	6	5	4	7
Machine M_2	4	5	3	2
Machine M_3	1	3	4	2
Machine M_4	2	4	5	1
Machine M_5	8	9	7	5

Also find the total elapsed time.

(Ans: Optimal sequence: 1 – 3 – 2 – 4; Minimum total elapsed time = 43 hours.)

3. Solve the following sequencing problem giving an optimal solution when no passing is allowede:

		Job				
		1	2	3	4	5
Machine	A	14	7	12	8	10
	B	5	6	4	7	3
	C	3	2	4	1	5
	D	10	12	8	15	16

(Ans: 2 – 4 – 5 – 1 – 3; Tmin = 76 time units; Idle times: 25, 51, 61, 15 time units respectively)

4. There are 4 jobs each of which has to go through the machines M_1, M_2, M_3, M_4, M_5 and M_6 in the order $M_1M_2M_3M_4M_5M_6$. Processing times in minutes are given. Determine the sequence of these four jobs which minimizes the total elapsed time T.

		Machines					
		M_1	M_2	M_3	M_4	M_5	M_6
Jobs	A	20	10	9	4	12	27
	B	19	8	11	8	10	21
	C	13	7	10	7	9	17
	D	22	6	5	6	10	14

(Ans: C – A – B – D; Tmin = 130 min)

Problems with 2 jobs through *M* Machines:

Consider two jobs, each of which is to be processed on M machines M_1, M_2 ……..... M_m in two different orders. The ordering of each of the two jobs through K machines is known in advance. Such ordering may not be the same for both the jobs. The exact or expected processing times on all the given machines are known.

Each machine can perform only one job at a time. The objective is to determine the optimal sequence of processing the jobs so as to minimize the total elapsed time.

The optimal sequence in this case can be obtained by making use of the graph. The procedure is given in the following steps.

Step 1 First draw a set of axes, where the horizontal axis represents processing time on job 1 and the vertical axis represents processing time on job 2.

Step 2 Mark the processing time for job 1 and job 2 on the horizontal and vertical lines respectively, according to the given order of machines.

Step 3 Construct various blocks starting from the origin (starting point), by pairing the same machines until the end point.

Step 4 Draw the line starting from the origin to the end point by moving horizontally, vertically and diagonally along a line which makes an angle of 45º with the horizontal line (base). The horizontal segment of this line indicates that the first job is under process while second job is idle. Similarly, the vertical line indicates that the second job is under process while first job is idle. The diagonal segment of the line shows that the jobs are under process simultaneously.

Step 5 An optimum path is the one that minimizes the idle time for both the jobs. Thus, we must choose the path on which diagonal movement is maximum.

Step 6 The total elapsed time is obtained by adding the idle time for either job to the processing time for that job.

Problems:

1. Use the graphical method to minimize the time needed to process the following jobs on the machines shown, i.e. for each machine find the job that should be done first. Also, calculate the total elapsed time to complete both jobs.

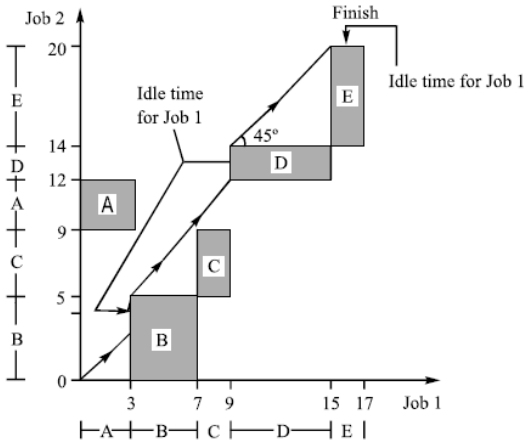
				Machine		
Job 1	Sequence:	A	B	C	D	E
	Time (hrs):	3	4	2	6	2
				Machine		
Job 2	Sequence:	B	C	A	D	E
	Time (hrs):	5	4	3	2	6

Solution:

The solution procedure for solving the above problem can be summarized in the following steps:

1. Draw a set of axes at right angle to each other where x-axis represents the processing time of job 1 on different machines while job 2 remains idle and y-axis represents processing time of job 2 while job 1 remains idle.

2. Mark the processing times for jobs 1 and 2 on x-axis and y-axis, respectively, according to the given order of machines as shown in Fig



Machine A takes 3 hours for job 1 and 3 hours for job 2. Construct the rectangle for machine A, as shown in above Fig. Similarly, construct other rectangles for machines B, C, D and E.

3. Construct various blocks starting from the origin by pairing the same machines until a point marked 'finished' is obtained.
4. Draw a line starting from the origin to the point marked 'finish' by moving horizontally, vertically and diagonally along a line that makes an angle of 45° with the horizontal axis. Moving horizontally along this line indicates that the first job is under process while the second job is idle. Similarly, moving vertically along this line indicates that the second job is under process while the first job is idle. The diagonal movement along this line shows that both the jobs are simultaneously under process.

   Since simultaneous processing of both jobs on a machine is not possible, therefore, a diagonal movement is not allowed. In other words, diagonal movement through rectangle areas is not allowed.
5. An optimal path is one that minimizes the idle time for both the jobs. Thus, we must choose the path on which diagonal movement is maximum as shown in Fig
6. Total elapsed time is obtained by adding the idle time for either job to the processing time for that job. In this example, the idle time for the chosen path is found to be 5 hrs and 2 hrs for jobs 1 and 2, respectively. The total elapsed time is calculated as follows:

   Elapsed time, Job 1 = Processing time of job 1 + Idle time for job 1
   = 17 + (2 + 3) = 22 hours

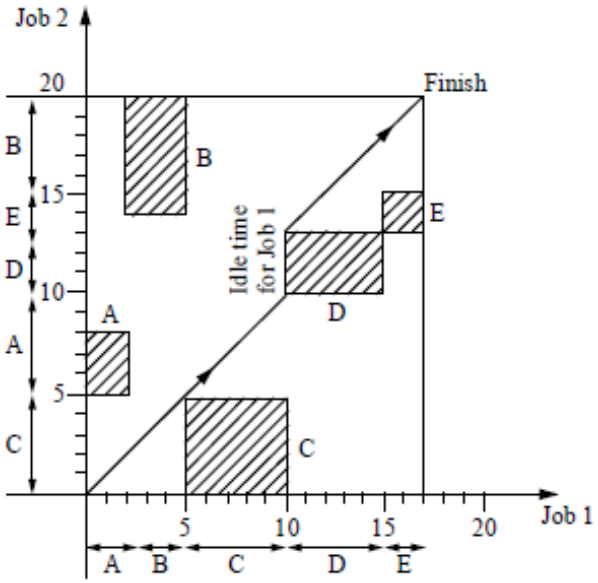
   Elapsed time, Job 2 = Processing time of job 2 + Idle time for job 2
   = 20 + (17 – 15) = 20 hours.

2. Use graphical method to minimize the time needed to process the following jobs on the machines shown below, i.e., for each machine find the job that should be done first. Also calculate the total time needed to complete both the jobs.

Job 1	Sequence of Machine:	*A*	*B*	*C*	*D*	*E*
	Time (hrs):	2	3	4	6	2
Job 2	Sequence of Machine:	*C*	*A*	*D*	*E*	*B*
	Time (hrs):	4	5	3	2	6

Solution:

The given information is shown in the figure. The shaded blocks represent the overlaps that are to be avoided.



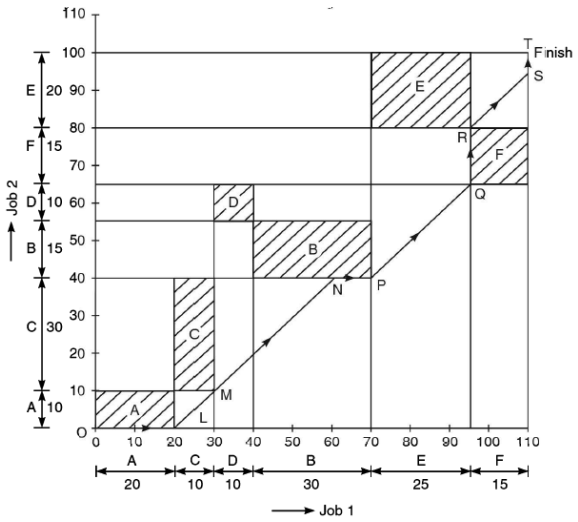
An optimal path is one that minimizes the idle time for job 1 (horizontal movement). Similarly, an optimal path is one that minimizes the idle time for job 2 (vertical movement).
For the elapsed time, we add the idle time for either of the job to the processing time for that job.
In this problem, the idle time for the chosen path is seen to be 3 hours for job 1 and zero for job 2.
Thus, the total elapsed time = 17 + 3 = 20 hours.

3. A workshop has six machines A. B, C, D, E and F. two jobs have to be processed through each pf these machines. The processing time on each machine and technological sequence of jobs is given below:
   Job 1: A – C – D – B – E – F
   Job 2: A – C – B – D – F – E

	Time in hours on machine					
	A	B	C	D	E	F
Job 1	20	30	10	10	25	15
Job 2	10	15	30	10	20	15

   In which order should the jobs be done on each of the machines to minimize the total time required to process the jobs? Also find the minimum elapsed time.

Solution:
Mark the processing times for job 1 and 2 on the horizontal and vertical axes according to the given technological ordering of the machines.
Construct the rectangular blocks by pairing the sane machine as shown in the fig. Now mark a path from the origin O to the point finish moving as much possible along 45^0 line.
Idle time for job 1 is 15 + 5 = 20 hours and for job 2 is 20 + 10 = 30 hours.



Total elapsed time = 110 + 20 = 130 hours (considering job1)
= 100 + 30 = 130 hours (considering job2)

4. Two major parts P_1 and P_2 for a product require processing through machine centres. The technological sequence of the parts on six machines and manufacturing times on each machine are

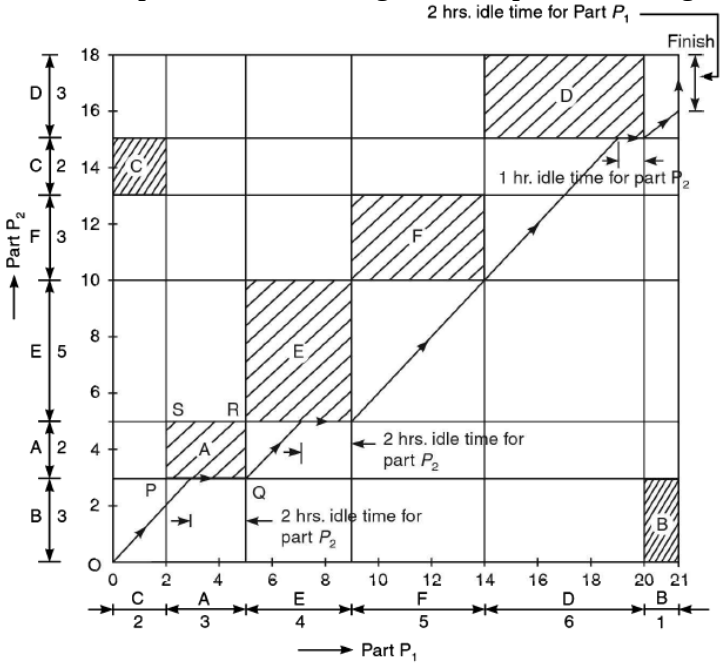
Part P_1	Machine sequence:	*C*	*A*	*E*	*F*	*D*	*B*
	Time (hrs):	2	3	4	5	6	1
Part P_2	Machine sequence:	*B*	*A*	*E*	*F*	*C*	*D*
	Time (hrs):	3	2	5	3	2	3

What would be the optimal scheduling to minimize the total processing time for these two parts? Find also the total elapsed time. For each machine specify the part that should be processed first.

Solution:

Mark the processing times for parts P_1 and P_2 on the horizontal and vertical axes according to the given technological ordering of the machines.

Construct the rectangular blocks by pairing the sane machine as shown in the fig. Now mark a path from the origin O to the point finish moving as much possible along 45^0 line.



Idle time for part P_1 is 2 hours and for part P_2 is 2 + 2 + 1 = 5 hours.

Total elapsed time = 21 + 2 = 23 hours (considering part P_1)

= 18 + 5 = 23 hours (considering part P_2)

Part that should be processed first:

Machine A: Part P_1 precedes part P_2
Machine B: Part P_2 precedes part P_1
Machine C: Part P_1 precedes part P_2
Machine D: Part P_1 precedes part P_2
Machine E: Part P_1 precedes part P_2
Machine F: Part P_1 precedes part P_2

5. Using the graphical method, calculate the minimum time needed to process jobs 1 and 2 on five machines A, B, C, D and E, i.e. for each machine find the job that should be done first. Also, calculate the total time needed to complete both jobs.

		Machine				
Job 1	Sequence:	A	B	C	D	E
	Time (hrs):	6	8	4	12	4
Job 2	Sequence:	B	C	A	D	E
	Time (hrs):	10	8	6	4	12

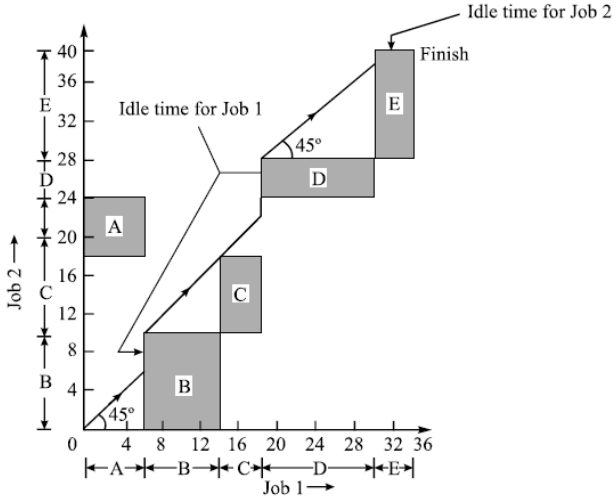
Solution:

Draw two axes at right angle to each other where x-axis represents the processing time of job 1 on different machines while job 2 remains idle and y-axis represents the processing time of job 2 on different machines while job 1 remains idle.

Mark the processing times for both the jobs 1 and 2 on x-axis and y-axis, respectively according to the given order of machines, as shown in Fig. For example, machine A takes 6 hours for job 1 as well as for job 2. Construct the rectangle for machine A as shown in Fig. 20.2. Similarly, construct other rectangles for machines B, C, D and E.

Draw a line starting from the origin to the point marked 'finished' by moving horizontally, vertically and diagonally along a line which makes an angle of 45° with the horizontal axis. A horizontal move represents processing of job 1 while job 2 remains idle; a vertical move represents processing of job 2 while job 1 remains idle. The diagonal movement (i.e. a 45° line) shows that both the jobs are under process simultaneously. Since simultaneous processing of both the jobs on a machine is not possible, therefore, diagonal movement is not allowed.

An optimal path is one that minimizes idle time for both jobs 1 and 2. This means that this path should coincide with 45° line to the maximum extent. Thus, we must choose the path on which diagonal movement is maximum, as shown in Fig. 20.2. The total elapsed time is obtained by adding idle time for either job to the processing time for that job.



The idle time for the chosen path is found to be 10 (= 4 + 6) hours for job 1 and 4 hours for job 2.

The total elapsed time is calculated as follows:

Elapsed time, Job 1 = Processing time + Idle time = 34 + (4 + 6) = 44 hours.
Elapsed time, Job 2 = Processing time + Idle time = 40 + (32 – 28) = 44 hours.

Practice Problems:

1. Two jobs are to be processed on four machines A, B, C and D. The technological order for these jobs on machines is as follows:

Job 1	A	B	C	D
Job 2	D	B	A	C

Processing times are given in the following table

	Machines			
	A	B	C	D
Job 1	4	6	7	3
Job 2	4	7	5	8

Find the optimal sequence of jobs on each of the machines.

(Ans: Idle time is 4 hours for job 1 and zero hour for job 2; Elapsed time for job 1 is 20 + 4 = 24.)

2. Use graphical method to minimize the time needed to process the following jobs on machines A, B, C, D and E. Find the total elapsed time to complete both jobs. Also find for each job, the machine on which it should be processed first.

		Machine				
Job 1	Sequence:	A	B	C	D	E
	Time (hrs):	2	3	5	2	1
Job 2	Sequence:	D	C	A	B	E
	Time (hrs):	6	2	3	1	3

(Ans: Tmin = 16 hrs; A(1), B(1), C(1), D(2), E(1 or 2))

3. A machine shop has four machines, A, B, C and D. Two jobs must be processed through each of these machines. The time (in hrs) taken on each of the machines and the necessary sequence of jobs through the shop are given below:

Job 1	Sequence:	A	B	C	D	E
	Time (hrs):	2	4	5	1	2
Job 2	Sequence:	D	E	A	C	B
	Time (hrs):	6	4	2	3	6

Use the graphic method to obtain the total minimum elapsed time.

(Ans: Total elapsed time is 15 hours.)

4. There are two jobs to be processed through five machines A, B, C, D and E. The prescribed technological order is

Job 1	A	B	C	D	E
Job 2	B	C	A	D	E

The process times in hours are given in table below

Job 1		Job 2	
Sequence of Machine	Time	Sequence of Machine	Time
A	3	B	5
B	4	C	4
C	2	A	3
D	6	D	2
E	2	E	6

Find out the optimal sequencing of the jobs on machines and minimum time required to process these jobs.

(Ans: Tmin = 22 hrs; A(1), B(2), C(2), D(1), E(2))